

Algorithmic Authority and Double Bayesian Distortion: How Generative AI Reshapes Evidence Sampling and Belief Updating under Uncertainty

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1. Introduction and Research Background

The rapid diffusion of generative artificial intelligence is changing the basic way human beings acquire information, organize evidence, and form judgments under uncertainty. Unlike search engines, databases, and traditional recommender systems, generative AI does not merely help users find information more quickly, nor does it simply recommend one option to them. It is increasingly used to interpret highly uncertain issues such as disease risk, investment prospects, career choice, public policy, legal responsibility, historical controversy, and social conflict. In such settings, users are not typically facing a factual query with a clearly existing answer; rather, they are facing a judgment task in which the evidence is incomplete, the sources are heterogeneous, explanations compete with one another, and conclusions are probabilistic.

In a traditional information environment, individuals usually encounter evidence first and then gradually form explanations and judgments. Even though search engines and recommender systems reduce the cost of obtaining information, users still need to compare evidence across multiple sources, identify conflicts, assess reliability, and decide when to stop searching. In other words, traditional information technology mainly changes the efficiency and path of information search rather than fully replacing individuals' direct contact with the evidence space. Although users may be influenced by ranking, recommendation, and attention-allocation mechanisms, they are still usually aware that they face an information environment composed of multiple sources, multiple viewpoints, and multiple pieces of evidence.

Generative AI changes this epistemic order. Large language models usually do not first present users with a set of scattered information sources. Instead, they directly generate a structured, coherent, and seemingly well-synthesized explanation. As a result, users often

encounter an algorithmically organized judgment framework before they have been fully exposed to the underlying evidence. This framework contains not only a conclusion, but also reasons, evidence ordering, conflict handling, and causal narrative. What users therefore face is no longer an unprocessed evidence environment, but an evidence environment that has already been pre-interpreted by an algorithm.

The key point of this change is that generative AI is not simply providing “more information.” It is completing in advance part of the epistemic work that users would otherwise have had to perform themselves. It helps users decide which evidence is more important, which conflicts can be interpreted as exceptions, which uncertainties can be compressed into understandable narratives, and which conclusions appear more reasonable under the current evidence. For users, the value of AI output comes not only from the amount of information it contains, but also from its formal completeness, fluency, and interpretability. A clearly structured, linguistically self-consistent, and logically coherent AI answer can give users the feeling that the main information has already been fully processed.

Accordingly, the social-scientific significance of generative AI should not be understood merely as a reduction in information-acquisition costs. More importantly, it may alter the basic process through which individuals form beliefs under uncertainty. In traditional belief formation, individuals usually confront the incompleteness, heterogeneity, and conflict of evidence first, and then gradually form a judgment within those uncertainties. In a generative AI environment, by contrast, individuals may first encounter a seemingly complete explanation and only then decide whether it is necessary to return to the evidence itself. In other words, cognitive closure may occur before users have adequately confronted evidentiary uncertainty.

From the perspective of behavioral economics and information economics, judgment under uncertainty contains at least two distinguishable stages. First, individuals decide whether to continue acquiring information. This stage involves the value of additional information, search costs, and optimal stopping. Second, individuals decide how to update beliefs after receiving new evidence. This stage involves evidentiary diagnosticity, prior beliefs, and Bayesian updating. The distinctive feature of generative AI is that it may intervene in both stages simultaneously. On the one hand, the AI’s synthetic answer may lead users to believe mistakenly that the main evidence has already been sufficiently covered, thereby reducing the perceived value of continued search. On the other hand, the AI’s coherent narrative may compress ambiguity and conflict within the problem, thereby reducing the degree to which users update on later disconfirming evidence.

This means that the risk of generative AI is not limited to whether it produces false information, biased content, or hallucinated facts. A subtler risk is that, even when the AI’s answer is not obviously wrong, its highly fluent and structurally complete explanatory form

may cause users to believe prematurely that they already understand the issue, or that the main evidence has already been adequately processed. The key questions then are no longer only “Did the user adopt an incorrect answer?” but also “Did the user stop searching for evidence too early?” and “Did the user underestimate the importance of later disconfirming evidence?”

Existing research on algorithmic judgment mainly focuses on whether humans trust, adopt, or avoid algorithmic advice, such as algorithm aversion, algorithm appreciation, and automation bias (Parasuraman and Riley, 1997; Lee and See, 2004; Dietvorst et al., 2015; Logg et al., 2019). This literature provides an important foundation for understanding human-machine decision-making, but it mostly studies traditional algorithmic advice, where the algorithm provides a predicted value, score, classification result, ranking result, or recommended option.

The difference between generative AI and traditional recommendation algorithms is not merely that the algorithm is more complex. More fundamentally, generative AI changes the form of information and the epistemic relationship users have to it. Traditional recommender systems mainly change the exposure structure of information through ranking, filtering, and personalized recommendation; their core function is to determine which discrete information items are more likely to enter the user’s field of vision (Resnick and Varian, 1997; Adomavicius and Tuzhilin, 2005; Bozdag, 2013). Users may be influenced by rankings and recommendations, but they still typically confront multiple relatively separate information entry points and must compare, interpret, and integrate them on their own.

Generative AI further changes the representational form of information. It does not simply arrange external information items, but directly generates a continuous, fluent, and structurally complete natural-language explanation. Because of mechanisms such as the Transformer architecture, autoregressive language modeling, instruction tuning, and reinforcement learning from human feedback, the typical outputs of generative AI often appear contextually relevant, semantically coherent, and “helpful” (Vaswani et al., 2017; Brown et al., 2020; Ouyang et al., 2022). This means that generative AI folds evidence selection, evidence weighting, conflict handling, and conclusion generation into a single answer, so that before users have been fully exposed to the underlying evidence, they are already facing a knowledge product that appears to have completed the synthesis.

Thus, the influence of generative AI lies not only in changing which information users see, but also in changing how users understand *how* the answer was generated. When facing a traditional recommendation algorithm, users are typically evaluating a ranking result. When facing generative AI, users are more likely to infer whether the AI has already understood the problem, seen enough evidence, covered the major dimensions, processed disconfirming evidence, and whether its summary represents the entire evidence environment. This is the

theoretical starting point for the concept of *perceived signal structure*.

To capture this mechanism, this proposal introduces the concept of perceived signal structure. Perceived signal structure refers to users’ subjective beliefs about the information-generation process behind a generative AI summary. More concretely, users do not merely assess whether the AI’s conclusion is credible; they also infer how much evidence the AI has observed, which dimensions of evidence it has covered, whether it has processed disconfirming information, whether it has integrated different sources, whether it has compressed evidentiary conflict, and whether the summary can represent a larger evidence pool. In other words, when users face AI output, they are not only receiving an answer; they are also forming a subjective model of the signal-generation process behind that answer.

This concept helps distinguish the present proposal from the existing literature on algorithmic trust. The influence of generative AI does not come entirely from “algorithmic authority,” nor does it arise solely from direct trust in the AI’s conclusion. Even if users do not fully believe the AI’s final judgment, they may still believe that the AI has already searched, filtered, and synthesized extensively. Even if users retain some doubts about the AI, they may still reduce the perceived necessity of continued search and the diagnosticity of later counterevidence because they believe the AI has already processed the main disconfirming evidence. The key influence of generative AI, therefore, is not simply to increase or decrease trust in the algorithm. Rather, it changes users’ understanding of the current evidence environment itself.

From the perspective of information economics, this process can be understood as a special kind of information-design effect. The classic framework of Bayesian persuasion shows that an information sender can influence a receiver’s posterior beliefs and action choices by designing the signal structure (Kamenica and Gentzkow, 2011). More broadly, the literature on information design studies how the designer of an information structure can shape receivers’ optimal behavior under uncertainty (Bergemann and Morris, 2019). This proposal draws on that line of thought but shifts the focus from an external sender designing a signal structure to users’ perceptions of the signal structure behind a generative AI summary.

Further, the literature on non-Bayesian persuasion shows that when receivers do not update according to the normative Bayesian rule, the effect of information design depends on the form of receivers’ systematic inferential biases (de Clippel and Zhang, 2022). This perspective is especially suitable for analyzing generative AI. AI does not merely transmit signals; in generating a summary, it reorganizes the signal structure. Users likewise do not merely update on a signal; before updating, they first form subjective inferences about the AI’s signal-generation process. The influence of generative AI therefore depends not only on the actual signal structure, but also on the signal structure that users perceive.

On this basis, the proposal develops a framework of “perceived signal structure and double

Bayesian distortion.” The core claim is that answer-first synthesis by generative AI may influence human belief formation under uncertainty through two mechanism paths.

First, at the information-acquisition stage, generative AI may induce an illusion of algorithmic exhaustiveness. Because AI summaries often appear in a structured, synthetic, and interpretive form, users may overestimate their evidentiary coverage, sample size, and representativeness. They may believe that the AI has already searched and integrated the major evidence sources, compared different viewpoints, and processed the key conflicts. As a result, the subjective value of additional search falls, the marginal return to continued information acquisition is underestimated, and individuals are more likely to stop evidence sampling too early.

Second, at the belief-updating stage, generative AI may produce an ambiguity-compression effect. The coherent narrative produced by AI tends to organize scattered, conflicting, and uncertain evidence into a clearer explanatory framework. When users believe that the AI has already processed the main disconfirming evidence, later counterevidence may be perceived as a signal with lower novelty, lower representativeness, or lower diagnosticity. As a result, users may encounter new disconfirming evidence without revising their prior beliefs to the degree warranted by its objective information content.

Generative AI may therefore create a *double Bayesian distortion*. The first distortion occurs at the information-acquisition stage, where users underestimate the value of additional signals and stop searching too early. The second distortion occurs at the information-processing stage, where users underestimate the diagnosticity of disconfirming signals and update too little. The term “double Bayesian distortion” here does not refer only to posterior updating errors in a narrow sense. Rather, it refers to simultaneous deviations from the normative Bayesian decision process in both information acquisition and belief updating.

The central theoretical argument of this proposal is that the key influence of generative AI does not lie only in changing what answer users accept. By providing a coherent explanation before users see the evidence, it changes the rules by which humans acquire information and update beliefs under uncertainty. It may lead users to search less because they believe the evidence space has already been adequately covered, and it may lead them to update less because conflicting evidence has already been compressed into an existing narrative. In this way, generative AI can produce a double Bayesian distortion under conditions of algorithmic authority, simultaneously altering both the information-acquisition and information-processing stages.

2. Literature Review

2.1 Information Acquisition and Evidence Sampling under Uncertainty

Belief formation under uncertainty does not begin with the final judgment; it begins with the decision of whether to keep looking for evidence. What individuals ultimately believe depends not only on how they interpret the evidence they have already seen, but also on how much evidence they saw before the judgment was formed, which evidence they saw, and at what point they stopped searching. Information acquisition is not a neutral preparatory step prior to belief formation. It is the first key site at which belief distortion can arise.

From a normative perspective, information search can be understood as a cost–benefit trade-off. When the expected value of additional information is higher than the cost of obtaining it, continuing to search is rational. When the marginal value of additional information is lower than the search cost, stopping is also rational. The literatures on rational inattention and costly information acquisition both emphasize that information is not free; individuals’ attention and search behavior should be understood as a process of information choice under cost constraints (Sims, 2003; Caplin and Dean, 2015).

Classic decision research likewise shows that people do not process all information under uncertainty costlessly and without limit. Instead, they adopt adaptive search strategies and decide when to stop searching based on task complexity, information cost, and expected return (Payne et al., 1993). In reality, however, stopping rules do not always conform to normative requirements. Individuals may seek supporting information because of confirmation bias, terminate search too early because of a need for cognitive closure, or reduce their willingness to continue exploring because the first information they receive appears too clear, satisfactory, or coherent (Nickerson, 1998; Lord et al., 1979).

Generative AI introduces a new question for research on information acquisition. When users see a highly structured, coherent, and seemingly well-synthesized AI summary before they encounter the underlying evidence, do they systematically overestimate the sufficiency of the current evidentiary state? Unlike a traditional search environment, an AI summary may make users feel that they have indirectly encountered a larger evidence space even though they have not actually seen the distribution, variance, and conflict structure of that evidence.

Accordingly, this proposal argues that AI may change users’ stopping rules by changing their subjective imagination of the evidence space. Users do not search less simply because they are “lazy.” They search less because they infer that the AI has already completed a considerable amount of information search, filtering, and integration. This mechanism is incorporated into the framework of perceived signal structure: if users believe that an AI

summary is based on more evidence, broader dimensions, and more representative evidence, then the subjective value of continued search will decline.

2.2 Belief Updating, Evidentiary Diagnosticity, and Under-Updating

After acquiring information, individuals still need to form or revise beliefs on the basis of the evidence they have obtained. In the normative Bayesian framework, individuals should adjust posterior beliefs according to the diagnosticity of new evidence. If a piece of evidence substantially raises or lowers the probability that a hypothesis is true, the individual's posterior judgment should change accordingly.

Yet a large body of behavioral decision research shows that human belief updating frequently deviates from normative Bayesian rules. Individuals may display conservatism and adjust too little in the face of new evidence; they may be influenced by anchoring so that later judgments depend too strongly on initial information; or they may reduce the weight of disconfirming evidence through motivated reasoning or confirmation bias. These distortions suggest that belief updating depends not only on the objective diagnosticity of the evidence itself, but also on how individuals understand the relationship between that evidence and their preexisting explanatory framework.

This point is especially important for the present proposal. Disconfirming evidence does not automatically change people's beliefs. It must first be recognized as evidence that is relevant, credible, and diagnostically meaningful before it can generate substantial updating. If individuals interpret disconfirming evidence merely as an exception, noise, a boundary case, or information already absorbed by the original narrative, then even objectively highly diagnostic evidence may not be given sufficient weight.

Existing research on belief updating has paid more attention to the effects of priors, motivation, cognitive ability, or information presentation format, and less attention to how the explanatory framework generated in advance by generative AI changes users' perceptions of the diagnosticity of later evidence. This proposal argues that this is the second key gap in understanding how AI affects human judgment. Generative AI may influence not only whether users continue to search for evidence, but also how they interpret the same evidence after they see it.

This motivates the mechanism of *ambiguity compression*. By providing a coherent, complete, and readable explanatory narrative, generative AI organizes evidence that was originally heterogeneous, conflicting, and probabilistic into a more stable framework of understanding. That framework may reduce users' perception of unresolved uncertainty and evidentiary conflict, making later disconfirming evidence seem less important, less representative, or less capable of changing the original judgment.

2.3 Algorithmic Advice, Algorithmic Trust, and Algorithmic Authority

Existing research on how humans respond to algorithmic outputs mainly revolves around algorithmic advice, algorithmic trust, algorithm aversion, algorithm appreciation, and automation bias. This research shows that humans neither consistently trust nor consistently reject algorithms. Sometimes, after seeing an algorithm make a mistake, people overavoid algorithmic judgment. At other times, they regard algorithms as more objective and more accurate than humans and therefore become more willing to adopt algorithmic advice. Research on automation bias further shows that, in some contexts, humans may overrely on automated systems and thereby neglect their own judgment or external evidence.

This literature provides an important basis for understanding human-machine decision relations, but it usually treats algorithmic output as a prediction, score, classification result, or recommended option. As a consequence, the main concern is often whether users adopt the algorithmic advice, revise their own judgment, overrely on the algorithm, or become averse to it after the algorithm errs. In short, the existing literature primarily focuses on the effect of algorithmic output on final judgment or choice.

The distinctiveness of generative AI lies not only in the fact that it provides explanations, but also in the fact that users may understand it as an epistemic agent with a knowledge state and judgmental capacity. Traditional recommendation algorithms are usually understood as ranking tools, filters, or decision aids. By contrast, generative AI interacts with users through natural-language dialogue and can explain, ask follow-up questions, revise, summarize, and respond to specific contexts. This interactive format makes users more likely to understand it as a quasi-subject that has already “seen,” “understood,” and “judged,” rather than as a neutral computational tool.

Research in human-computer interaction has long shown that humans apply social rules and social expectations to computer systems. The CASA paradigm argues that when computers display language, responsiveness, interactivity, or social cues, people unconsciously treat them as social actors (Reeves and Nass, 1996; Nass and Moon, 2000). Research in social cognition likewise shows that people anthropomorphize and attribute minds to entities on the basis of interactivity, goal directedness, and behavioral complexity, inferring what they “know,” what they have “understood,” and whether they have “already considered” particular information (Epley et al., 2007; Waytz et al., 2010).

This point is especially important in the context of generative AI. Woodruff and Hewitt (2026) argue that, although LLMs can generate fluent and seemingly knowledgeable answers, their outputs do not satisfy knowledge conditions such as reflective regulation, resistance to revision, and normative commitment. They therefore should not be designed or understood

as epistemic authority, but rather as knowledge-building partners (Woodruff and Hewitt, 2026). Qi and Pan (2026) similarly argue in the context of evidence-based medicine that general-purpose LLMs can generate medically plausible evidence-related outputs, yet remain limited in source verifiability, methodological rigor, and responsibility attribution (Qi and Pan, 2026).

Accordingly, the “algorithmic authority” discussed in this proposal is not just the general belief that “the AI answer is correct.” It refers, more specifically, to the user’s tendency to regard AI as an epistemic agent that has already carried out the work of evidence search, evidence synthesis, and preliminary judgment on the user’s behalf. When facing generative AI, users may ask not only “Is this answer right?” but also infer “Has it already seen enough evidence?”, “Has it already considered the opposing view?”, and “Has it already completed the synthesis for me?” These inferences are precisely what this proposal calls perceived signal structure.

The contribution of the present project to the literature on algorithmic advice is therefore as follows: existing studies mainly explain whether humans accept algorithmic output, whereas this proposal focuses on how algorithmic authority changes human judgments about evidentiary sufficiency and evidentiary diagnosticity.

2.4 Generative AI, Information Design, and Non-Bayesian Persuasion

The literature on information design studies how a sender who controls the presentation of information can shape a receiver’s beliefs and actions by designing the signal structure. In the classic Bayesian persuasion framework, the sender does not directly control the receiver’s action but instead chooses an information structure that shapes the receiver’s posterior beliefs and thereby influences action choice (Kamenica and Gentzkow, 2011). The later literature on information design extends this idea into a unified framework for analyzing the relationship among information structures, posterior beliefs, and behavioral responses (Bergemann and Morris, 2019). However, this traditional framework usually assumes that receivers can interpret signals correctly according to Bayesian rules. In reality, individuals often display systematic deviations in probabilistic inference. Non-Bayesian persuasion therefore emphasizes that when the receiver is not a Bayesian updater, the actual posterior can be understood as the result of applying some distortion function to the normative Bayesian posterior (de Clippel and Zhang, 2020).

This proposal builds on that idea and extends it into the context of generative AI. Generative AI is not a simple extension of traditional recommendation algorithms. Search engines, recommender systems, and news-feed algorithms mainly change users’ information-exposure

structures through ranking, filtering, and personalization; their core function is to determine which discrete information items are more likely to enter the user’s field of vision (Resnick and Varian, 1997; Adomavicius and Tuzhilin, 2005; Bozdag, 2013). In such systems, users are still usually facing multiple relatively separate information entry points, such as links, products, articles, videos, or ratings, and must compare, interpret, and integrate them on their own. In short, traditional recommendation algorithms mainly change what users “see” and what they “see first.”

The crucial difference with generative AI is that it further changes both the expressive form of information and the epistemic relationship to it. Rather than arranging external information items, it directly generates a continuous, fluent, and structurally complete natural-language explanation. Because of mechanisms such as the Transformer architecture, autoregressive language modeling, instruction tuning, and reinforcement learning from human feedback, the typical outputs of generative AI often appear contextually relevant, semantically coherent, natural in tone, and “helpful” (Vaswani et al., 2017; Brown et al., 2020; Ouyang et al., 2022). What users face, therefore, is not a set of information sources waiting for further evaluation, but a synthetic answer that has already been organized, compressed, and narrativized by the model. Generative AI folds into a single output what were traditionally scattered across multiple cognitive stages: evidence selection, evidence weighting, conflict handling, causal explanation, and conclusion generation.

This gives generative AI a persuasive structure that differs from that of traditional recommendation algorithms. Traditional recommender systems may influence information exposure, but they still generally preserve visible separation among multiple information items. Generative AI instead integrates heterogeneous evidence, conflicting information, and uncertainty into a single, outwardly coherent explanatory narrative. Narrative transportation theory shows that coherent narratives can influence belief by absorbing attention, emotion, and imagination, thereby lowering the tendency to rebut or evaluate critically (Green and Brock, 2000). In the generative AI context, this mechanism may be amplified further: what AI outputs is not scattered evidence, but an explanatory framework that has already completed an initial act of meaning-making. Users may therefore more readily understand it as a stable posterior judgment rather than as a hypothesis that still requires testing.

Yet the formal coherence of generative AI is not equivalent to real understanding or adequate evidence synthesis. Large language models can generate highly fluent text, but linguistic fluency does not guarantee semantic grounding, factual sensitivity, or evidence-sensitive judgment. This is precisely the basis of the critique advanced by Bender et al. (2021): language models operate mainly at the level of linguistic form, while humans easily infer meaning understanding and knowledge processing from fluent syntax and coherent discourse (Bender et al., 2021). The risk of generative AI, then, is not only that it may generate false content, but that users may misread its fluency and completeness as signs that the system has already

adequately processed the underlying evidence.

More than that, users may understand generative AI as an epistemic agent with its own knowledge state and judgmental capacity. Traditional recommendation algorithms are usually viewed as back-end rankers, filters, or auxiliary tools. Generative AI, by contrast, responds in natural language, can explain, revise, summarize, acknowledge uncertainty, and adjust answers to context. Research in human–computer interaction shows that when computer systems display language, responsiveness, and social cues, humans tend to apply social rules to them and treat them as social actors (Reeves and Nass, 1996; Nass and Moon, 2000). Social cognition research likewise shows that people anthropomorphize and attribute mentality on the basis of interactivity, goal directedness, and behavioral complexity. In the generative AI context, these tendencies create fertile ground for users to infer more from the AI’s answer than the answer itself contains.

2.5 Gaps in the Literature and the Positioning of This Proposal

Taken together, the existing literature provides an important foundation for the present proposal, but it still leaves an insufficiently explained theoretical gap: existing studies have rarely analyzed how generative AI changes users’ subjective understanding of *how* signals are generated and, in turn, how that understanding affects information acquisition and belief updating.

First, the literature on information acquisition and evidence sampling shows that whether individuals continue searching depends on their judgments about the value of additional information, search cost, and the sufficiency of current evidence. Yet this literature generally focuses on users’ own search costs, attentional limits, or motivational biases, and pays less attention to whether users overestimate the sufficiency of the current informational state when generative AI provides a synthetic explanation before they have contacted the original evidence. In other words, the existing literature has not adequately explained whether users reduce the subjective value of continued search because they believe the AI has already covered and integrated the main evidence.

Second, the literature on belief updating shows that people may display conservative updating, confirmation bias, or low weighting of disconfirming evidence, and that the subjective diagnosticity of evidence affects posterior revision. Yet this literature rarely examines how the coherent explanatory framework generated in advance by generative AI changes users’ understanding of subsequent evidence. In particular, when users believe that the AI has already considered counterevidence, later disconfirming information may no longer be seen as new and highly diagnostic evidence, but as a local exception that has already been absorbed by the original narrative.

Third, the literature on algorithmic advice and algorithmic trust mainly focuses on whether users adopt, revise, or avoid algorithmic output, while continuing to interpret algorithmic output as a prediction, score, recommendation, or classification. The distinctive feature of generative AI is that it not only gives a conclusion, but also organizes evidence, explains causal relations, handles conflict, and generates coherent narratives in natural language. The key question is therefore not merely whether users believe the AI’s answer, but how they understand the evidentiary basis of that answer: Has the AI seen enough evidence? Has it covered the main dimensions? Has it processed disconfirming information? Can its summary represent the overall evidence environment?

Fourth, the emerging literature on generative AI has already begun to address bias, hallucination, automation bias, narrative persuasion, and illusions of understanding, but it still needs to unpack the behavioral mechanisms behind those phenomena more directly. Existing research shows that AI outputs can affect judgment, but it rarely distinguishes between two different sources of deviation: does final judgment become distorted because users acquired less evidence earlier, or because they later assigned lower weight to the same evidence? This distinction is crucial both for understanding how AI affects belief formation and for designing appropriate governance interventions.

The present proposal is developed around this gap. It does not treat generative AI merely as a tool that provides advice. Rather, it treats generative AI as a meaning-making system that changes both users’ signal environment and their understanding of signals. It introduces the concept of perceived signal structure to describe users’ subjective inferences about the evidence-generation process behind an AI summary, including perceived sample size, coverage breadth, representativeness, degree of disconfirming-evidence processing, and compression loss. The argument is that answer-first synthesis by generative AI may affect both stages simultaneously: at the information-acquisition stage, users may overestimate the evidentiary coverage and representativeness of the AI summary and therefore reduce the subjective value of continued search; at the information-processing stage, users may believe that disconfirming evidence has already been considered or absorbed by the AI and therefore reduce the perceived diagnosticity of later counterevidence.

The question this proposal asks, then, is not whether AI affects human judgment in some general sense. It asks a more specific mechanism question: how does generative AI, by changing users’ understanding of the signal-generation process, affect their judgments of evidentiary sufficiency and evidentiary diagnosticity under uncertainty, and how does that in turn produce insufficient evidence sampling and insufficient belief updating?

3. Theoretical Framework

3.1 Generative AI as an Algorithmic Meaning-Making Agent

This proposal treats generative AI as a special kind of information designer. Its difference from traditional recommendation algorithms lies not simply in whether it filters information, but in the fact that it generates a seemingly complete explanatory structure in continuous natural language. Traditional recommendation algorithms mainly alter the ordering and visibility of discrete information items. Generative AI, by contrast, compresses evidence selection, evidence weighting, conflict handling, and conclusion generation into a single answer, so that users confront an “already interpreted” information environment before they ever see the raw evidence.

More importantly, the conversational interface of generative AI induces users to make inferences about the system’s knowledge state. Users do not merely see an answer; they may believe that the answer rests on extensive evidence processing, cross-dimensional comparison, and absorption of counterevidence. The informational effect of generative AI therefore lies not only in changing what users see, but also in changing what users believe the system has already seen, understood, and synthesized.

In traditional belief formation, the usual order is:

$$\text{Evidence} \rightarrow \text{Explanation} \rightarrow \text{Judgment}.$$

In a generative AI environment, the order is closer to:

$$\text{AI Judgment} \rightarrow \text{AI Explanation} \rightarrow \text{Optional Evidence Verification}.$$

This reversal in sequence is the key to the present theoretical framework. Users may encounter what looks like a complete posterior judgment first and only afterward decide whether to continue engaging with the underlying evidence. As a result, cognitive closure may occur before uncertainty has been fully confronted.

3.2 Algorithmic Authority and the Perception of Evidentiary Sufficiency

This proposal defines *algorithmic authority* as users’ subjective perception that AI has relatively strong capability in information search, evidence integration, and judgment generation. Algorithmic authority is not the same as simple trust. Simple trust asks whether “the AI’s answer is correct.” Algorithmic authority asks whether “the AI has already processed the

evidence environment sufficiently on my behalf.”

This distinction is crucial. Even if users do not fully believe the AI’s final conclusion, they may still believe that the AI has already retrieved and integrated the main information. In that case, the influence of AI may not appear as direct adoption of the answer, but as reduced willingness to continue searching. In other words, algorithmic authority first affects users’ judgments about evidentiary sufficiency.

In decision-making under uncertainty, whether individuals continue to search for information depends on their assessment of the value of additional information. If they believe continued search might substantially alter their judgment, continued sampling is reasonable. If they believe the existing information is already sufficient, stopping is also reasonable. The problem is that a synthetic AI output may cause users to overestimate the sufficiency of the current informational state and thereby reduce the subjective value of additional information too early.

Accordingly, the core function of algorithmic authority is to alter users’ answers to the following questions:

- Do I still need more evidence?
- Is the current answer already comprehensive enough?
- Is continued search still worth it?

This means that the influence of AI is not simply to push users toward one specific answer. Rather, it changes their judgment about whether the evidence is already sufficient. It affects the stopping rule in the process of belief formation, not merely the final choice.

3.3 Path One: The Illusion of Algorithmic Exhaustiveness and Distortion in Information Acquisition

The first mechanism path occurs at the information-acquisition stage. This proposal defines the *illusion of algorithmic exhaustiveness* as users’ belief that the answer generated by AI has already covered, filtered, and integrated a sufficiently broad evidence space, even though the AI may not actually have exhausted the relevant evidence.

This illusion does not necessarily arise from blind trust in AI. It may arise from the formal features of AI output itself. Generative AI answers are often clearly structured, linguistically fluent, complete in argument, and inclined to present multiple factors in a summarized way. Precisely this form of completeness makes it easy for users to feel that the main information has already been processed.

The illusion of algorithmic exhaustiveness lowers users’ perceived value of additional information. Users may believe that continuing to search will bring only repetitive evidence or evidence with low marginal value, rather than evidence that could substantially change the current judgment. As a result, they become more likely to stop searching early, sample fewer pieces of evidence, and move into final decision-making more quickly.

This mechanism chain can be summarized as follows:

AI answer-first synthesis → algorithmic authority → illusion of algorithmic exhaustiveness
→ decline in perceived value of additional information → premature search stopping
→ endogenous undersampling of evidence.

This path explains how AI may cause users to “see less.”

It is important to emphasize that such behavior need not take the form of subjective laziness or overt irrationality. If users believe that the AI has already completed a sufficiently comprehensive integration of information, then reducing search may appear reasonable from their perspective. The real problem is that the “formal completeness” of AI output may be misread as “evidentiary completeness.”

3.4 Path Two: Ambiguity Compression and Distortion in Information Processing

The second mechanism path occurs at the information-processing stage. This proposal defines *ambiguity compression* as the process by which AI converts heterogeneous, conflicting, and probabilistic evidence into a highly coherent explanatory narrative, thereby reducing users’ perception of unresolved uncertainty and evidentiary conflict.

Complex problems usually admit multiple possible explanations. Evidence may be incomplete, internally conflicting, or point toward different conclusions under different conditions. Traditionally, such ambiguity reminds users to keep their judgment open and to revise it when new evidence appears. Generative AI, however, often tends to produce answers that are clear, readable, and coherent. In order to generate a “helpful” answer, AI may place conflicting evidence inside a unified narrative, explain counterexamples as boundary conditions or exceptions, and treat them as secondary factors.

For example, the AI may say:

Although some counterexamples exist, the overall trend still supports this judgment.

Although the evidence is not entirely consistent, the more reasonable explanation is ...

These differences mainly arise from contextual factors and do not alter the core conclusion.

Such formulations are not necessarily wrong, but they change users’ psychological perception of conflicting evidence. Evidence that might otherwise have strong diagnostic force may be reinterpreted as a local anomaly, background noise, or information already absorbed by the master narrative.

As a result, AI may influence not only how much evidence users see, but also how they interpret the evidence they do see. Even if users later encounter new disconfirming information, they may underestimate the extent to which that information challenges their prior judgment, producing insufficient belief updating.

This mechanism chain can be summarized as follows:

AI answer-first synthesis → coherent explanatory narrative → ambiguity compression
→ reduced perceived diagnosticity of disconfirming evidence → insufficient belief updating
→ deviation from Bayesian updating.

This path explains how AI may cause users to “update less.”

3.5 Double Bayesian Distortion

The central theoretical concept of this proposal is *double Bayesian distortion*. This concept refers to the possibility that generative AI creates simultaneous distortions at two stages of human belief formation: insufficient sampling at the information-acquisition stage and insufficient updating at the information-processing stage.

The first distortion is a distortion in information acquisition. In normative terms, a rational decision-maker should continue searching for evidence whenever the expected value of additional information exceeds the search cost. If the AI’s synthetic answer causes users to believe mistakenly that the evidence is already sufficient, and they stop searching before acquiring enough information, a distortion arises on the information-acquisition side.

The second distortion is a distortion in information processing. In the normative Bayesian framework, after observing new evidence, individuals should revise posterior beliefs according to that evidence’s diagnosticity. If later disconfirming evidence has high diagnostic value,

but users under-update because they previously accepted the AI’s coherent explanation, a distortion arises on the information-processing side.

The theoretical significance of double Bayesian distortion lies in the fact that it distinguishes between two error sources that are often conflated:

- Error type one: users did not acquire enough evidence.
- Error type two: users acquired the evidence but did not give it enough weight.

The first is an information-acquisition problem; the second is an information-processing problem. Both can produce distorted posterior beliefs and lower decision quality, but they involve different mechanisms and therefore require different governance responses. If the problem is undersampling, the intervention should encourage continued search or reveal the incompleteness of the evidence space. If the problem is under-updating, the intervention should heighten the salience and diagnosticity of conflicting evidence.

The theoretical framework of the present proposal can therefore be summarized as follows:

Answer-first synthesis by generative AI ↓

Algorithmic authority ↓

Information-acquisition path	Information-processing path
Illusion of algorithmic exhaustiveness	Ambiguity compression
→ lower perceived value of additional information	→ lower diagnosticity of disconfirming evidence
→ premature stopping of sampling	→ insufficient belief updating

⇓

Double Bayesian distortion

⇓

Posterior belief distortion / decline in decision performance.

4. Research Questions

The overarching research question of this proposal is:

How does answer-first synthesis by generative AI reshape the process of human belief formation under uncertainty by changing evidence sampling, ambiguity perception, and belief updating?

More specifically, the proposal asks the following questions:

RQ1: Does answer-first synthesis by generative AI change users’ perceived signal structure?

This question concerns how users understand the evidence-generation process behind an AI summary. The key issue is not whether users fully believe the AI’s final conclusion, but whether they believe the AI has seen more evidence, covered more evidentiary dimensions, processed the main disconfirming evidence, and generated a synthesis that can represent the overall evidence environment.

RQ2: Does perceived signal structure reduce the subjective value of continued search?

This question concerns the psychological mechanism at the information-acquisition stage. If users believe that the AI summary has already adequately covered and integrated the relevant evidence, they may believe that continued search would bring only repetitive or low-marginal-value information and therefore reduce the perceived value of additional evidence.

RQ3: Does generative AI produce undersampling of evidence through perceived signal structure?

This question concerns the core behavioral outcome. If users in the AI answer-first condition voluntarily sample less evidence, stop searching earlier, and this effect is explained by perceived signal structure and perceived information value, then generative AI is not merely affecting final judgment; it is changing users’ rules for information acquisition.

5. Research Hypotheses

5.1 The Perceived Signal Structure Path

H1: Compared with traditional search-style or evidence-list-style information presentation, AI answer-first increases inflation in users’ perceived signal structure.

More specifically, users in the AI answer-first condition will believe that the summary is based on more evidence, broader evidentiary dimensions, and a more representative informational basis, and they will be more likely to believe that the AI has already processed the main disconfirming evidence.

H2: Inflation in perceived signal structure lowers the perceived informational value of continued search.

The more users believe that the AI summary has already sufficiently covered and represented the overall evidence pool, the more likely they are to regard the marginal value of drawing additional evidence as low.

H3: AI answer-first leads users to reduce the number of evidence samples they voluntarily draw.

Compared with evidence-list, search-style, or no-summary conditions, users in the AI answer-first condition will voluntarily sample fewer additional pieces of evidence and stop information search earlier.

H4: Perceived signal structure and perceived information value mediate the effect of AI answer-first on evidence undersampling.

AI answer-first does not simply make users “too lazy to search.” Rather, by changing users’ subjective understanding of the coverage, representativeness, and degree of disconfirming-evidence processing behind the AI summary, it lowers the perceived necessity of continued search and thereby produces undersampling.

The mechanism chain is:

AI answer-first → inflation in perceived signal structure
→ decline in perceived value of additional information → decline in voluntary sample size.

5.2 The Signal-Structure Calibration Path

H5: Signal-structure calibration prompts can weaken the effect of AI answer-first on undersampling.

If the AI summary explicitly discloses that its evidentiary coverage is limited, does not represent the full evidence set, or warns that disconfirming evidence may remain unprocessed, then inflation in users’ perceived signal structure will be weaker, the perceived value of continued search will be higher, and voluntary sample size will also increase.

The mechanism chain is:

signal-structure calibration prompt → reduced inflation in perceived signal structure
→ higher perceived value of additional information → larger voluntary sample size.

5.3 Exclusion Tests

H6: The effect of AI answer-first on undersampling cannot be explained entirely by general algorithmic trust or anchoring on the initial answer.

After controlling for users’ trust in AI accuracy, their confidence in the initial conclusion, and the direction and strength of the initial answer, perceived signal structure should still significantly predict the perceived value of continued search and the number of evidence samples drawn voluntarily.

6. Core Theoretical Contributions

This proposal seeks to contribute to the existing literature on AI and human judgment in three ways.

First, it reconceptualizes generative AI from “algorithmic advice” to an “algorithmic meaning-making agent.” Existing research often focuses on whether users adopt algorithmic output, whereas the present project emphasizes that the influence of generative AI arises at an earlier stage of problem understanding and evidence organization. Generative AI does not merely provide an answer. It organizes evidence in advance, compresses conflict, and generates an interpretive framework for the user.

Second, the proposal decomposes the influence of AI on human judgment into two stages: information acquisition and information processing. By introducing the mechanisms of the illusion of algorithmic exhaustiveness and ambiguity compression, it distinguishes between two different deviations: users “seeing less” and users “updating less.” This distinction helps avoid reducing the effect of AI to trust, laziness, anchoring, or automation bias alone.

Third, the proposal introduces the theoretical framework of double Bayesian distortion and embeds the behavioral effects of generative AI inside the normative frameworks of information acquisition, optimal stopping, and Bayesian updating. This framework helps explain not only how AI changes individual belief formation, but also how its potential welfare consequences and governance responses should be evaluated.

In short, the key influence of generative AI is not merely that it changes which answer users accept. It changes how users judge whether evidence is sufficient, whether conflict matters, and whether beliefs need revision. Generative AI is transforming humans from agents who form judgments on the basis of evidence into agents who manage evidence on the basis of algorithmic explanation. The present proposal studies the behavioral mechanisms and decision consequences of this shift in cognitive order.

9. Research Design Overview

This proposal contains three experimental studies.

Study	Goal	Core Design	Core Contribution
Study 1	Test double Bayesian distortion	Two-stage sequential sampling + yoked control + logit updating	Distinguish acquisition from processing
Study 2	Identify AI-specific mechanisms	Structural manipulation of exhaustiveness cues and ambiguity compression	Test the illusion of algorithmic exhaustiveness and ambiguity compression
Study 3	Test governance mechanisms and net payoff	Governance-treatment experiment under varying cost structures	Evaluate which interventions improve judgment quality without imposing excessive friction

10. Study 1: Main Two-Stage Sequential Sampling Experiment

10.1 Research Objective

Study 1 tests whether generative AI produces double Bayesian distortion:

- Stage one: does AI lead users to stop evidence sampling too early?
- Stage two: does AI lead users to under-update, in log-odds terms, on highly diagnostic disconfirming signals?

Study 1 is divided into Study 1a and Study 1b.

- Study 1a: natural sequential sampling, used to estimate the total effect of AI on the full belief-formation process.

- Study 1b: a yoked control design, used to identify the direct effect of processing bias after controlling for information-set size.

10.2 Experimental Scenario

Preferred scenario:

Judgment of corporate default risk.

Reasons:

- the scenario allows a natural prior probability to be set;
- it allows positive and negative evidence to be designed;
- each piece of evidence can be assigned a likelihood ratio;
- participants can be asked to report probabilities;
- investment / hedging decisions can be designed;
- token incentives can be used to calculate net payoff.

Task example:

Participants are asked to judge whether a given firm will enter a high-default-risk state within the next year. The system provides a large research-report database. Participants may pay to inspect evidence and must ultimately report the probability that the firm belongs to the high-default-risk type.

This proposal uses:

an evidence pool that is fully controlled by the researcher but appears open-ended to the participant.

The participant sees:

You can never prove to reviewers what the “objective strength” of a natural-language cue such as “executive resignation” really is, because natural language contains tremendous semantic ambiguity, and different participants may map the same text into very different prior beliefs. If the objective normative baseline itself is not calculable, then every subsequent measure of “deviation” becomes a castle in the air.

Academic precedent.

This design is not only preceded in the literature; it belongs to a very large and classic research tradition: the sequential sampling paradigm and active information acquisition.

- Rooted in psychology and decision science: classic urn experiments (Edwards, 1965).
- Information-systems and process research: the classic Mouselab paradigm (Payne et al., 1993), where information is hidden and participants must click to reveal it, allowing researchers to measure search cost and stopping points.
- Modern macro/behavioral economics frontier: information-choice experiments under rational inattention, such as Caplin and Dean (2015) and Enke and Zimmermann (2019).

These top-journal designs succeed not by having participants read long reports, but by following one core rule: strip away uncontrolled semantics and move to a data-generating process (DGP).

How to quantify evidentiary strength: the breakthrough rule.

To realize a design that is completely controlled by the researcher but still appears open-ended to participants, “textual reports” must be reduced into structured signals with explicit likelihood ratios (LRs).

In the instructions, participants should be shown the conditional probability matrix of the signal-generating process. This is the only way to ensure that all participants face the same objective likelihood function.

Concrete illustration:

- Do *not* say: “You may search the database for research reports.”
- Instead say: “The database contains 50 independent automated risk-control indicators (for example, a cash-flow early-warning system and supply-chain public-opinion monitoring). Historical data show that these indicators have imperfect diagnostic accuracy.”

Then explicitly specify $P(\text{signal} \mid \text{state})$:

- If the firm’s true state is **high default risk**, then any indicator drawn from the database shows a “red light” (negative) with probability 0.7, and a “green light” (positive) with probability 0.3.
- If the firm’s true state is **low default risk**, then any indicator drawn from the database shows a “red light” with probability 0.3, and a “green light” with probability 0.7.

At that point, the strength of each piece of evidence is perfectly quantified:

$$LR_{\text{negative}} = \frac{P(\text{red light} \mid \text{high risk})}{P(\text{red light} \mid \text{low risk})} = \frac{0.7}{0.3} = 2.33$$

Likewise, the strength of drawing a green light is:

$$LR_{\text{positive}} = \frac{P(\text{green light} \mid \text{high risk})}{P(\text{green light} \mid \text{low risk})} = \frac{0.3}{0.7} = 0.43$$

The strongest implementable version of Study 1.

To give this mathematical model the ecological validity of an “AI research-report” environment, the system should add a layer of user-interface packaging.

Underlying logic (fully controlled).

The researcher pre-specifies the true state (for example, the firm is actually low risk). The generated red-light and green-light indicators in the participant’s sampling pool strictly follow the probability distribution above. The entire Bayesian posterior path created by each draw is known to the researcher in the background. The researcher can calculate precisely, given a token cost c , what the optimal stopping point N^* should be.

Surface presentation (apparently open).

When the participant clicks “draw evidence,” the system does not merely display a “red light.” Instead, it displays a short structured excerpt.

If the internal code generates a red light, the interface may display:

Indicator #12 (liquidity stress test): failed. Detail: AI analysis indicates that short-term debt-servicing capacity is in the historical low range. (Historical accuracy parameter: 70%.)

If the internal code generates a green light, the interface may display:

Indicator #45 (core-asset pledge ratio): passed. Detail: AI analysis indicates that collateral value remains stable, with no abnormal impairment observed. (Historical accuracy parameter: 70%.)

Treatment for the AI synthesis group.

Before the participant enters the sampling pool, the AI first appears and says:

Based on a rapid full scan of the 50 underlying indicators in the database, this AI judges that the firm faces a relatively high risk of default. Although some local indicators (such as core assets) remain acceptable, the dominant trend (such as liquidity conditions) has deteriorated severely.

The key point is that the AI text is highly coherent (high ambiguity compression) and implicitly suggests that it has already seen all 50 “balls” (the illusion of algorithmic exhaustiveness).

How advisors and reviewers would evaluate this design.

Once this DGP is written into the proposal, the central concerns of reviewers and advisors are greatly reduced:

- Endogeneity is addressed: all evidence LRs are pre-specified mathematical facts.
- A gold-standard benchmark is obtained: because the LR is fixed, an absolute Bayesian posterior curve can be drawn, and participants’ subjective reports can be directly compared against it to produce a rigorous measure of Bayesian deviation.
- Experimental tension is preserved: participants still feel that they are browsing a large financial database, which preserves business realism, while the back end remains an extremely clean statistical urn.

Thus, the participant experiences:

- a large research-report database;
- a large amount of evidence related to the focal firm;
- one piece of evidence revealed with each click;
- a small token cost for each revelation;
- greater reward for more accurate final judgment.

The researcher controls in the background:

- the evidence-generating distribution;
- the direction of each piece of evidence;
- the likelihood ratio of each piece of evidence;

- local variance across evidence items;
- the LR of the high-diagnostic exogenous shock;
- the optimal sample size N^* ;
- the normative Bayesian posterior probability.

This preserves both:

- ecological validity: users believe they face an open information environment;
- internal validity: the researcher knows the evidence structure and the normative Bayesian benchmark.

10.4 Study 1a: Natural Sequential Sampling

Experimental conditions

Condition	Description	Function
AI answer-first	Participants first see an AI synthetic judgment and then sample freely	Main treatment
Human expert answer-first	Participants first see a human expert's synthetic judgment and then sample freely	Distinguish AI-specific effects from structured synthesis
Search-only free	No synthetic judgment; free sampling only	Natural search baseline
AI answer-last	Participants sample freely first and see the AI synthesis only later	Distinguish answer-first ordering from mere AI exposure

Stage one: endogenous sampling

Process:

1. Provide the base event and prior probability.
2. Depending on condition, display an AI synthesis, a human-expert synthesis, or neither.
3. The user enters the evidence pool.

4. Each viewed piece of evidence costs tokens.
5. The user may stop at any time.
6. After stopping, the user reports an *interim belief*.

Core measures:

- N_{actual} : actual voluntary sample size
- stopping point
- search cost
- perceived VOI: perceived value of continued information search
- algorithmic exhaustiveness: the illusion of algorithmic exhaustiveness
- interim belief: interim posterior probability
- interim confidence

Core dependent variable:

$$\text{Sampling Deviation} = N^* - N_{\text{actual}}$$

where N^* is the normatively optimal sample size given search cost and error loss.

If

$$N_{\text{actual}} < N^*,$$

then endogenous undersampling is present.

Stage two: exogenous shock

After participants voluntarily stop and report their interim belief, the system forcibly presents one or more highly diagnostic pieces of disconfirming evidence.

Key design features:

- all participants see the same type of shock;
- the LR of the shock is preset by the researcher;
- the shock is conditionally independent of the stage-one sampled signals;
- each participant's own interim belief is used as the prior;

- normative updating is computed in log-odds form.

Process:

1. The user has already reported an interim belief.
2. The system pushes a high-LR disconfirming shock.
3. The user reports a final belief.
4. Logit updating deviation is calculated.
5. The user makes a final decision.
6. Net payoff is calculated from decision accuracy, search cost, and error loss.

10.5 Study 1b: Yoked Control Processing Test

Research objective

Study 1a estimates the total effect of AI on the final outcome, but different sampling amounts in stage one may influence stage-two processing. To identify processing bias cleanly, Study 1b controls information-set size.

Condition design

Condition	Description
AI answer-first	Participants first see the AI synthetic judgment and then sample freely
Search-only yoked	Participants do not see the AI, but are assigned the same number of evidence items as the AI group

Logic of the yoked control.

If a participant in the AI group stops at

$$N = 2,$$

then the matched participant in the yoked search-only group also sees only

$$N = 2$$

pieces of evidence before entering the same exogenous-shock stage.

This ensures that the AI group and the yoked group enter stage two with the same information-set size.

If the AI group still shows weaker log-odds updating under that condition, then the difference is not simply because the AI group “read less,” but is more likely to arise from the effect of the early AI narrative on evidence-processing weights.

Implementation options

Two approaches are possible.

Option 1: real-time matching.

Participants in the yoked group are matched in real time to the actual stopping points of AI-group participants.

- Advantage: the strictest form of individual-level matching.
- Disadvantage: experimentally more complex to execute.

Option 2: pilot-based distribution matching.

Use a pilot study to estimate the AI group’s distribution of sample sizes, then make the yoked group passively receive evidence according to that distribution.

- Advantage: stable execution.
- Disadvantage: not fully individualized.

The proposal favors a pilot-based yoked distribution, with individual matching used as a robustness check.

10.6 Core Indicators in Study 1

Information-acquisition stage

- Voluntary Sample Size = N_{actual}
- Sampling Deviation = $N^* - N_{\text{actual}}$
- Search Cost

Information-processing stage

The main outcome uses log-odds updating deviation.

$$\text{logit}(P) = \ln\left(\frac{P}{1-P}\right)$$

Actual update:

$$\text{Actual Logit Update}_i = \text{logit}(P_{\text{final},i}) - \text{logit}(P_{\text{interim},i})$$

Normative update:

$$\text{Normative Logit Update} = \log(LR_{\text{shock}})$$

Updating deviation:

$$\text{Updating Deviation}_i = |[\text{logit}(P_{\text{final},i}) - \text{logit}(P_{\text{interim},i})] - \log(LR_{\text{shock}})|$$

Implied weight:

$$w_i = \frac{\text{logit}(P_{\text{final},i}) - \text{logit}(P_{\text{interim},i})}{\log(LR_{\text{shock}})}$$

Interpretation:

- $w_i = 1$: the shock is fully absorbed according to the Bayesian rule;
- $0 < w_i < 1$: the weight of the shock is underestimated;
- $w_i \approx 0$: almost no updating;
- $w_i < 0$: updating occurs in the wrong direction.

Final-outcome stage

- Final Bayesian Deviation
- Decision Accuracy
- Net Payoff = Accuracy Reward – Search Cost – Error Loss

11. Study 2: Mechanism-Decomposition Experiment for AI-Specific Effects

11.1 Research Objective

Study 2 identifies two AI-specific mechanisms directly:

- the illusion of algorithmic exhaustiveness, which affects information acquisition;
- ambiguity compression, which affects information processing.

The core principle is to avoid low-level declarative manipulations and instead use structural cue manipulations. In other words, participants are not directly told that “AI searched many sources” or “AI only saw a few sources.” Different perceptions are induced naturally through the structure of the answer itself.

11.2 Experimental Design

The study uses a 2×2 design.

Factor	Level 1	Level 2
Exhaustiveness cue	high exhaustiveness	low exhaustiveness
Ambiguity compression	high compression / absorbed counterevidence	low compression / suspended counterevidence

The four conditions are:

1. high exhaustiveness $\times$ high compression
2. high exhaustiveness $\times$ low compression
3. low exhaustiveness $\times$ high compression
4. low exhaustiveness $\times$ low compression

All conditions hold constant:

- the final conclusion;

- the evidence pool;
- the direction of the conclusion;
- task difficulty;
- and, as far as possible, text length.

11.3 Structural Manipulation of Exhaustiveness Cues

High-exhaustiveness version

Wide evidentiary coverage is implied through cross-dimensional integration:

Drawing on financial statements, industry conditions, supply-chain data, management turnover, the macro interest-rate environment, peer default patterns, and expert-interview information, the firm appears likely to enter a high-default-risk state within the next year.

Low-exhaustiveness version

Narrower coverage is implied through single-dimension analysis:

Based on recent financial statements, cash flow, leverage, and debt-servicing indicators, the firm appears likely to enter a high-default-risk state within the next year.

The final conclusion is identical in both versions. The only difference is the breadth of information coverage that users infer from the answer.

11.4 Structural Manipulation of Ambiguity Compression: Absorption vs. Suspension

The proposal no longer relies on a coarse manipulation of “with conclusion” versus “without conclusion.” Instead, it uses:

- high compression: present disconfirming evidence but explain it away;
- low compression: present disconfirming evidence but leave its conflict unresolved.

High ambiguity compression: explain-away / subsume

Overall, the firm’s default risk appears high. Although the firm recently obtained a new round of financing, this financing mainly consists of high-interest short-term bridge funding, which itself suggests substantial liquidity pressure. Combined with worsening cash flow, concentrated near-term debt maturities, and revenue volatility, this financing signal does not weaken the high-default-risk judgment, but instead further supports the interpretation of rising financial stress.

Low ambiguity compression: suspend

Overall, the firm’s default risk appears high, but this judgment rests on a clear tension within the evidence. Worsening cash flow and concentrated near-term debt maturities support the high-risk judgment, whereas recent financing success constitutes a countervailing signal. The current model places more weight on the cash-flow deterioration signal, but the meaning of the financing stability has not yet been fully clarified. If later information shows that this financing is stable over the long term, the default-risk judgment may need to be revised downward.

Both versions present the same dominant conclusion:

default risk is high.

The difference is that:

- the high-compression version absorbs the counterevidence into the main narrative;
- the low-compression version preserves the counterevidence as a conflict not yet fully resolved.

11.5 Measures in Study 2

Mechanism variables

- illusion of algorithmic exhaustiveness
- perceived breadth of information coverage
- perceived VOI of continued search
- perceived evidentiary sufficiency

- perceived evidentiary conflict
- perceived ambiguity compression
- perceived diagnosticity of the reverse signal
- self-consistent cognitive closure

Behavioral variables

- voluntary sample size
- tokens willing to pay for additional evidence
- whether the participant chooses to inspect opposing evidence
- logit updating deviation
- implied shock weight w_i
- final net payoff

11.6 Expected Results

- high exhaustiveness cues → stronger illusion of algorithmic exhaustiveness → lower perceived VOI → less sampling;
- high ambiguity compression → stronger self-consistent cognitive closure → lower perceived diagnosticity of reverse signals → weaker log-odds updating;
- high exhaustiveness × high ambiguity compression → the strongest double Bayesian distortion and the lowest net payoff.

12. Study 3: Governance Mechanisms and Net Payoff Experiment

12.1 Research Objective

Study 3 tests whether different governance mechanisms can reduce double Bayesian distortion and whether those mechanisms are practically viable.

The key question is not:

Which mechanism maximizes verification?

It is:

Which mechanism can improve judgment quality without imposing excessive cost or causing users to abandon the system, and thereby improve net payoff?

12.2 Experimental Conditions

Condition	Description
Baseline AI answer-first	AI directly gives a synthetic answer by default
Conflict-first AI	AI presents the key conflict first and then gives a conclusion
Uncertainty-calibrated AI	AI gives a probability range, evidentiary limits, and uncertainty reminders
Source-before-answer	users must inspect a minimum amount of evidence before seeing the AI answer
Optional paid verification	users may pay for verification and receive higher reward for correct judgment
Mandatory verification	users must inspect several pieces of evidence before making a decision

12.3 Loss Function and Net Payoff

The design uses real token incentives:

- a certain number of tokens is deducted for each piece of evidence viewed;
- more accurate final judgments receive higher rewards;
- incorrect decisions incur token losses;
- final tokens are converted into real compensation.

Core formula:

$$\text{Net Payoff} = \text{Accuracy Reward} - \text{Search Cost} - \text{Error Loss}$$

To improve external validity, Study 3 does not rely on only one cost structure. Instead, it manipulates different decision environments.

Environment	Verification Cost	Error Cost	Real-World Analogue
Low-risk environment	high	low	everyday information search
Medium-risk environment	medium	medium	organizational judgment
High-risk environment	low	high	finance, medicine, safety
Time-pressure environment	high time cost	high error cost	crisis decision-making

This allows the proposal to test:

- in which contexts light-touch governance works;
- in which high-risk contexts mandatory verification is worthwhile;
- whether users abandon the system because of governance friction.

12.5 User Choice and Retention

In the final round of tasks, users are allowed to choose freely among systems:

1. direct-answer system
2. conflict-first system
3. uncertainty-calibrated system
4. source-before-answer system
5. mandatory-verification system

The following are measured:

- system choice
- willingness to continue using the system
- realized net payoff
- subjective burden

- task completion time
- user attrition rate

This responds directly to the real deployment problem: if a governance mechanism improves accuracy but causes substantial user attrition, then its practical governance value is limited.

13. Empirical Strategy

13.1 Identifying Distortion in Information Acquisition

Main model:

$$N_{\text{actual},i} = \beta_0 + \beta_1 \text{AIAnswerFirst}_i + \beta_2 \text{HumanExpert}_i \\ + \beta_3 \text{AIAnswerLast}_i + \beta_4 \text{Controls}_i + \varepsilon_i$$

Mediation model:

$$\text{AI answer-first} \rightarrow \text{Algorithmic Exhaustiveness} \rightarrow \text{Perceived VOI} \rightarrow N_{\text{actual}}$$

Core test:

Does AI answer-first reduce voluntary sample size through the illusion of algorithmic exhaustiveness?

13.2 Identifying Distortion in Information Processing

Main outcome:

$$\text{Updating Deviation}_i = |[\text{logit}(P_{\text{final},i}) - \text{logit}(P_{\text{interim},i})] - \log(LR_{\text{shock}})|$$

Model:

$$\text{UpdatingDeviation}_i = \beta_0 + \beta_1 \text{AIAnswerFirst}_i + \beta_2 \text{HumanExpert}_i \\ + \beta_3 \text{SearchOnly}_i + \beta_4 \text{logit}(P_{\text{interim},i}) \\ + \beta_5 \text{Controls}_i + \varepsilon_i$$

In Study 1b, after adding yoked control:

$$\text{UpdatingDeviation}_i = \beta_0 + \beta_1 \text{AIAnswerFirst}_i + \beta_2 \text{YokedSearch}_i \\ + \beta_3 \text{InformationSetSize}_i + \beta_4 \text{Controls}_i + \varepsilon_i$$

Key comparison:

AI answer-first vs. search-only yoked

If the AI group still under-updates, then processing bias cannot be explained entirely by differences in information-set size.

13.3 Estimating Implied Weights

Computation:

$$w_i = \frac{\text{logit}(P_{\text{final},i}) - \text{logit}(P_{\text{interim},i})}{\log(LR_{\text{shock}})}$$

Model:

$$w_i = \beta_0 + \beta_1 \text{AIAnswerFirst}_i + \beta_2 \text{AmbiguityCompression}_i \\ + \beta_3 \text{Controls}_i + \varepsilon_i$$

Expected result:

In the AI answer-first condition and in the high-ambiguity-compression condition, w_i should be lower.

13.4 Distinguishing Belief from Preference

To avoid conflating belief effects with preference effects, the proposal simultaneously measures:

- subjective probability reports
- information-purchase behavior
- final choice
- risk preference
- ambiguity aversion
- confidence calibration
- net payoff

If AI mainly changes risk preference, then probability reports, information purchase, and logit updating need not move consistently. If AI changes belief updating, then these indicators should jointly exhibit systematic deviation.

13.5 Handling Double Counting

This proposal explicitly models the AI summary as:

a lossy compression of the underlying evidence environment

and not as:

an independent new source of evidence.

Accordingly, the normative Bayesian model does not simply add the AI summary as an independent signal. The AI summary is only a compressed presentation of the underlying evidence pool. The value of continued sampling lies in recovering what the AI compression process may have omitted:

- local variance
- disconfirming evidence
- evidence heterogeneity
- alternative explanations
- tensions in evidentiary weighting

Theoretical statement:

Generative-AI synthesis should not be modeled as independent evidence; it should be modeled as a lossy compression of the evidence environment. Because the compression process may suppress local variance and the salience of conflicting evidence, a rational Bayesian decision-maker should continue sampling whenever the expected value of restoring omitted variance exceeds the marginal search cost.

14. Variables and Measures

14.1 Independent Variables

- AI answer-first synthesis
- human-expert answer-first synthesis

- search-only free
- search-only yoked
- AI answer-last
- high vs. low exhaustiveness cue
- high vs. low ambiguity compression
- conflict-first design
- uncertainty-calibrated design
- source-before-answer
- mandatory verification

14.2 Mediators

- illusion of algorithmic exhaustiveness
- perceived breadth of information coverage
- perceived VOI of continued search
- perceived evidentiary sufficiency
- ambiguity compression
- perceived evidentiary conflict
- perceived diagnosticity of the reverse signal
- self-consistent cognitive closure

14.3 Evidence-Acquisition Outcomes

- voluntary sample size
- stopping point
- search cost
- sampling deviation
- deviation from the optimal sample size N^*

14.4 Evidence-Processing Outcomes

- $\text{logit}(P_{\text{final}}) - \text{logit}(P_{\text{interim}})$
- logit updating deviation
- implied shock weight w_i
- belief revision after the reverse signal

14.5 Decision Outcomes

- final posterior probability
- final choice
- judgment accuracy
- net payoff
- system choice
- willingness to continue using the system
- user attrition rate

15. Expected Contributions

15.1 From AI Anchoring to Dual Bayesian Distortion

This proposal no longer reduces the influence of AI to a simple anchoring effect. Instead, it argues that AI may produce two different kinds of Bayesian distortion:

- undersampling at the information-acquisition stage;
- insufficient log-odds updating at the information-processing stage.

This shifts the project from traditional psychological-bias research toward normative belief-updating research.

15.2 From Advice-Taking to Algorithmic Sensemaking

The proposal does not treat AI as an ordinary source of advice, but as a meaning-making agent.

The function of AI is not merely to say:

I recommend that you choose A.

Rather, it first organizes the world for the user:

- which evidence matters most;
- which conflicts can be absorbed;
- which conclusions are most reasonable;
- whether further search is still necessary.

This extends the literature on AI advice-taking.

15.3 Proposing the Illusion of Algorithmic Exhaustiveness as an AI-Specific Mechanism

The proposal argues that one of the key differences between AI and human experts is that users may believe AI covers a larger information space.

This illusion of algorithmic exhaustiveness lowers the perceived informational value of continued search and thereby triggers the stopping rule earlier.

15.4 Proposing Ambiguity Compression as an Information-Processing Mechanism

The proposal argues that AI does not merely provide answers. It explains, absorbs, and re-constructs conflicting evidence, compressing a complex evidence environment into a coherent narrative.

This ambiguity compression lowers the perceived diagnosticity of counterevidence and may produce attenuation in log-odds updating when users later encounter a high-LR shock.

15.5 Recasting AI Governance as a Net-Benefit Problem

The proposal does not simply argue that “more verification is better.” Instead, it evaluates governance mechanisms within a net-benefit framework.

The core claim is:

- optimal governance is not the maximization of verification;
- optimal governance balances accuracy gains, search costs, error losses, and user retention.

16. Anticipated Critiques and Responses

Critique 1: Is this just confirmation bias in new packaging?

Response:

The proposal acknowledges that confirmation bias may be present, but its contribution lies in identifying AI-specific antecedent conditions. By comparing AI answer-first, human-expert answer-first, search-only free, and search-only yoked conditions, the study can distinguish:

- a general structured-synthesis effect
- versus an AI-specific algorithmic-authority effect.

If equally coherent human-expert reports produce the same effect, then the mechanism is mainly structured synthesis. If the AI condition is stronger, and the difference is explained by the illusion of algorithmic exhaustiveness, then there is an AI-specific component.

Critique 2: Is the smaller update in the AI group just a probability-boundary effect?

Response:

The proposal does not use absolute probability differences as the main outcome. It uses log-odds updating:

$$\text{Updating Deviation} = |[\text{logit}(P_{\text{final}}) - \text{logit}(P_{\text{interim}})] - \log(LR_{\text{shock}})|$$

On the log-odds scale, Bayesian updating is linear, which avoids ceiling effects caused by probability boundaries.

Critique 3: Is under-updating in the AI group only because the group saw less in stage one?

Response:

Study 1b uses a search-only yoked control to ensure that the AI group and the yoked group possess the same amount of evidence before entering the exogenous-shock stage.

If the AI group still shows weaker log-odds updating under equal information-set size, then the processing distortion cannot be explained entirely by differences in reading volume.

Critique 4: If AI explain-away is correct, is lowering the weight of disconfirming evidence not rational?

Response:

The proposal does not claim that every explain-away move is wrong. What it tests is whether, when objectively highly diagnostic disconfirming signals exist, AI's coherent narrative causes users to underestimate the normative evidentiary weight of those signals.

Critique 5: Are token incentives sufficient to offset cognitive laziness?

Response:

The proposal does not assume that token incentives fully eliminate cognitive laziness. On the contrary, cognitive-effort cost is part of the theory itself. Study 3 manipulates incentive strength and cost structure to test whether the AI effect persists even in high-incentive environments.

Possible incentive levels:

- low incentive
- medium incentive
- high incentive

If the AI effect persists under high incentives, then it is not merely a low-effort compliance effect.

Critique 6: Does discrete sampling sacrifice ecological validity?

Response:

Discrete sampling is used in order to obtain a computable Bayesian benchmark. A fully naturalistic long-text search environment has higher ecological validity, but it is difficult to determine the LR and normative posterior associated with each signal.

The proposal therefore adopts a two-layer strategy:

- main experiment: discrete signal sampling, to ensure internal validity and a normative benchmark;
- supplementary experiment: long-form evidence packets or a simulated web environment, to test external validity.

17. Limitations

First, although the experimental evidence pool appears open-ended to participants, it is still a simulated environment constructed by the researcher. It cannot fully reproduce the complexity of real search environments that contain webpages, long texts, videos, and social information flows.

Second, the Bayesian benchmark in this proposal depends on likelihood ratios and evidence-generation structures specified by the researcher. Although this helps identify normative deviation, it also means that the conclusions must be further validated in other settings.

Third, AI-specific mechanisms may correlate with general trust, cognitive laziness, and obedience to authority. The proposal will reduce confounding through control variables, mediation models, and comparison groups, but it cannot exhaust every possible psychological mechanism at once.

Fourth, token incentives in the experiment cannot be fully equivalent to the large gains and losses present in high-stakes real-world decisions. Study 3 alleviates this issue by manipulating the cost structure, but external validity will still require later field experiments or platform data.

Fifth, in the real world, AI explain-away of disconfirming evidence may sometimes be high-quality reasoning. The present proposal tests only whether, in an experimental environment where the objective LR is known, AI narrative causes users to underestimate highly diagnostic reverse signals.

18. Governance Implications

The governance implication of this proposal is not simply to tell users to “verify more.” Instead, it is to design different information architectures for different risk environments.

Low-risk environments

Examples:

- everyday information search
- low-cost consumer decisions

Suitable mechanisms:

- lightweight uncertainty reminders
- optional source expansion
- low-friction verification prompts

Not suitable:

- mandatory verification
- source-before-answer

The reason is that the cost may exceed the benefit.

Medium-risk environments

Examples:

- organizational judgment
- ordinary investment analysis

Suitable mechanisms:

- conflict-first AI
- uncertainty-calibrated AI
- source-traceable summary

The goal is to reduce ambiguity compression and excessive cognitive closure while preserving usability.

High-risk environments

Suitable mechanisms:

- source-before-answer

- mandatory verification
- confidence calibration
- required exposure to disconfirming evidence

In environments with high error costs, the net benefit of strong governance may be positive.

Core governance principles

1. One should not optimize only for answer fluency.
2. AI should not overcompress evidentiary conflict.
3. High-risk systems should explicitly present key disconfirming evidence.
4. Source transparency is not enough; conflict transparency is also needed.
5. Optimal governance does not maximize verification; it maximizes net benefit.
6. AI systems should evaluate their effects on users' search-stopping rules and updating weights.

19. Expected Findings

This proposal expects to find the following:

First, AI answer-first will reduce users' willingness to continue sampling, especially when users form a strong illusion of algorithmic exhaustiveness.

Second, AI answer-first will cause users to stop searching earlier, and their voluntary sample size may fall below the normatively optimal sample size.

Third, even when using a log-odds updating metric, the AI answer-first group may still discount the weight of highly diagnostic disconfirming signals.

Fourth, in the yoked control, even after holding information-set size constant, the AI answer-first group may still show greater updating deviation, indicating that AI narrative affects not only information acquisition but also information processing.

Fifth, the negative effects of AI will not appear in every setting. In low-risk, low-error-cost tasks, AI may improve efficiency. But in high-risk, high-error-cost tasks, the undersampling and under-updating induced by AI are more likely to create net welfare losses.

Sixth, the effectiveness of governance mechanisms depends on cost structure. Conflict-first and uncertainty-calibrated designs may improve belief updating while imposing relatively low burden. Mandatory verification may be more valuable in high-risk settings, but in low-risk settings it may reduce net benefit and user retention because its cost is too high.

References

- Adomavicius, G. and Tuzhilin, A. (2005). Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, 17(6):734–749.
- Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623. ACM.
- Bergemann, D. and Morris, S. (2019). Information design: A unified perspective. *Journal of Economic Literature*, 57(1):44–95.
- Bozdag, E. (2013). Bias in algorithmic filtering and personalization. *Ethics and Information Technology*, 15(3):209–227.
- Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., and Amodei, D. (2020). Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33*, pages 1877–1901.
- Caplin, A. and Dean, M. (2015). Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*, 105(7):2183–2203.
- de Clippel, G. and Zhang, X. (2020). Non-bayesian persuasion. Working paper version dated March 4, 2020.
- de Clippel, G. and Zhang, X. (2022). Non-bayesian persuasion. *Journal of Political Economy*, 130(10):2594–2642.
- Dietvorst, B. J., Simmons, J. P., and Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1):114–126.

- Edwards, W. (1965). Optimal strategies for seeking information: Models for statistics, choice reaction times, and human information processing. *Journal of Mathematical Psychology*, 2(2):312–329.
- Enke, B. and Zimmermann, F. (2019). Correlation neglect in belief formation. *The Review of Economic Studies*, 86(1):313–332.
- Epley, N., Waytz, A., and Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4):864–886.
- Green, M. C. and Brock, T. C. (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, 79(5):701–721.
- Kamenica, E. and Gentzkow, M. (2011). Bayesian persuasion. *American Economic Review*, 101(6):2590–2615.
- Lee, J. D. and See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1):50–80.
- Logg, J. M., Minson, J. A., and Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151:90–103.
- Lord, C. G., Ross, L., and Lepper, M. R. (1979). Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of Personality and Social Psychology*, 37(11):2098–2109.
- Nass, C. and Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1):81–103.
- Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2):175–220.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askeel, A., Welinder, P., Christiano, P., Leike, J., and Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*, pages 27730–27744.
- Parasuraman, R. and Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2):230–253.
- Payne, J. W., Bettman, J. R., and Johnson, E. J. (1993). *The Adaptive Decision Maker*. Cambridge University Press, Cambridge.

- Qi, W. and Pan, L. (2026). Epistemic and ethical limits of large language models in evidence-based medicine: From knowledge to judgment. *Frontiers in Digital Health*, 7:1706383.
- Reeves, B. and Nass, C. (1996). *The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places*. CSLI Publications and Cambridge University Press, Stanford, CA.
- Resnick, P. and Varian, H. R. (1997). Recommender systems. *Communications of the ACM*, 40(3):56–58.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems 30*, pages 5998–6008.
- Waytz, A., Cacioppo, J., and Epley, N. (2010). Who sees human? the stability and importance of individual differences in anthropomorphism. *Perspectives on Psychological Science*, 5(3):219–232.
- Woodruff, E. and Hewitt, J. (2026). Epistemic agency in the age of large language models: Design principles for knowledge-building ai. *AI*, 7(3):99.